

Symmetric Key Crypto

Block Ciphers



(Iterated) Block Cipher

- Plaintext and ciphertext consists of fixed sized blocks
- Ciphertext obtained from plaintext by iterating a **round function**
- Input to round function consists of key and the output of previous round
- Usually implemented in software

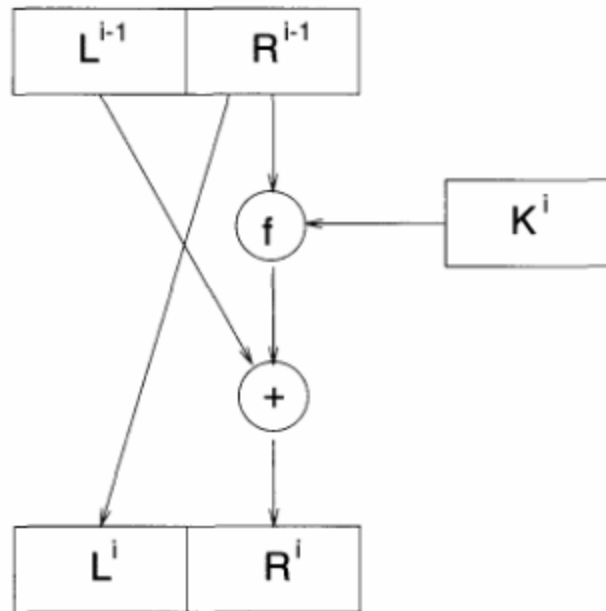
Feistel Cipher

- Named after Horst Fiestel
- **Feistel cipher** refers to a type of block cipher design, not a specific cipher
- Split plaintext block into left and right halves: Plaintext = (L_0, R_0)
- For each round $i=1,2,\dots,n$, compute
$$L_i = R_{i-1}$$
$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$
where F is **round function** and K_i is **subkey**
- Ciphertext = (L_n, R_n)

Feistel Cipher

- Decryption: Ciphertext = (L_n, R_n)
- For each round $i=n, n-1, \dots, 1$, compute
$$R_{i-1} = L_i$$
$$L_{i-1} = R_i \oplus F(R_{i-1}, K_i)$$
where F is round function and K_i is subkey
- Plaintext = (L_0, R_0)
- Formula "works" for any function F
- But only secure for certain functions F

Encryption



Repeating/ Iterating this transformation we obtain the Feistel Cipher

Data Encryption Standard

- ❑ DES developed in 1970's
- ❑ DES was developed by National Bureau of Standards
- ❑ Based on IBM Lucifer cipher
- ❑ U.S. government standard
- ❑ Most widely used block cipher in world

DES

- ❑ DES uses the two basic techniques of cryptography - confusion and diffusion.
- ❑ At the simplest level, diffusion is achieved through numerous permutations and confusions is achieved through the XOR operation and the S-Boxes.
- ❑ This is also called an S-P network.

□ As there are over 70,000,000,000,000,000 (seventy quadrillion) possible keys of 56 bits, the feasibility of deriving a particular key in this way is extremely unlikely in typical threat environments.

Introduction

- The algorithm is designed to encipher and decipher blocks of data consisting of 64 bits under control of a 64-bit key
- A block to be enciphered is subjected to an initial permutation IP and then to a complex key-dependent computation
- Finally the permutation which is the inverse of the initial per(IP^{-1})

56 bit key from 64 bit key

- ❑ Discard every 8th bit from the original key

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

16

17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

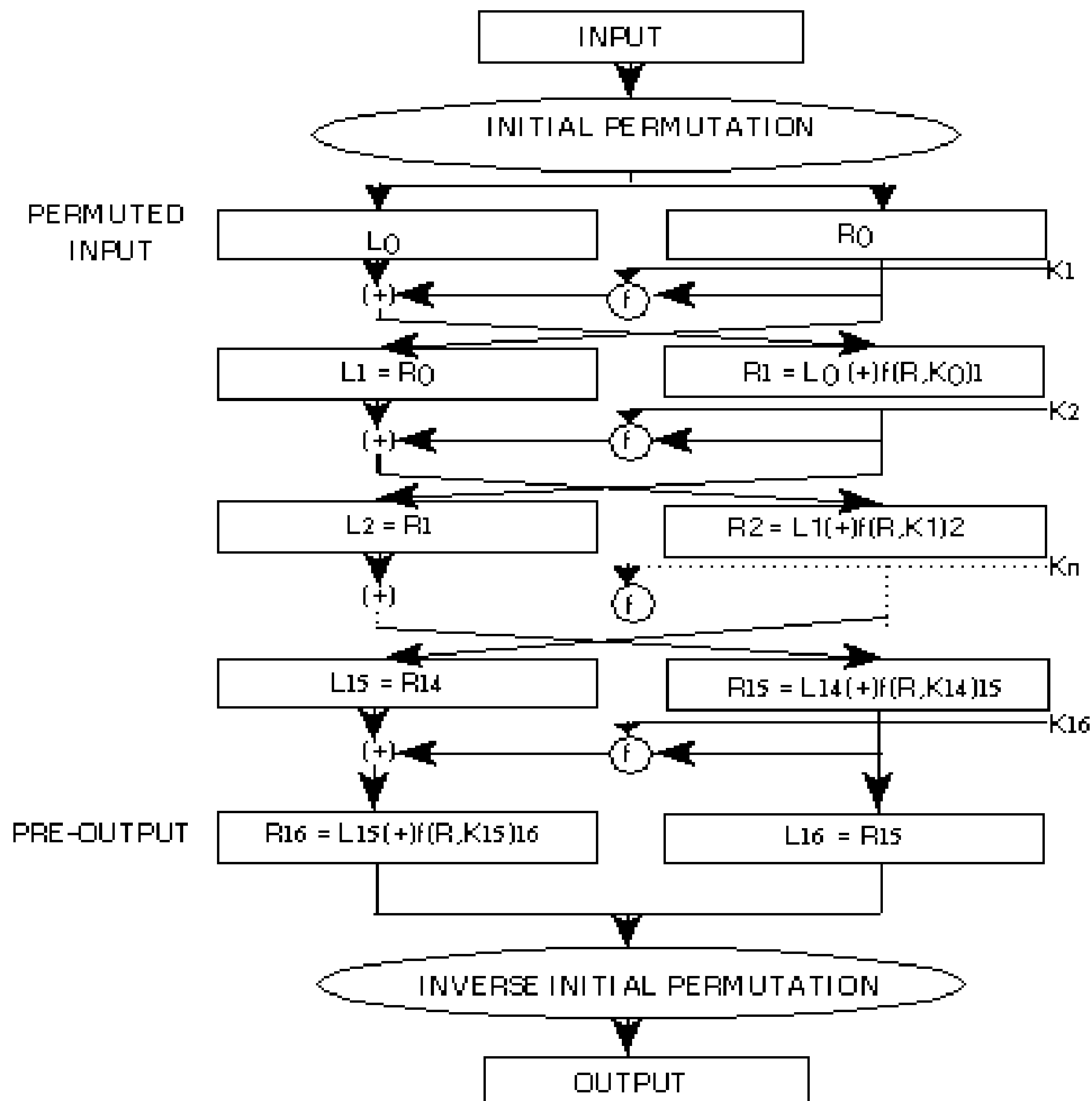
32

33 34 35 36 37 38 39 40 41 42 43 44 45 46 47

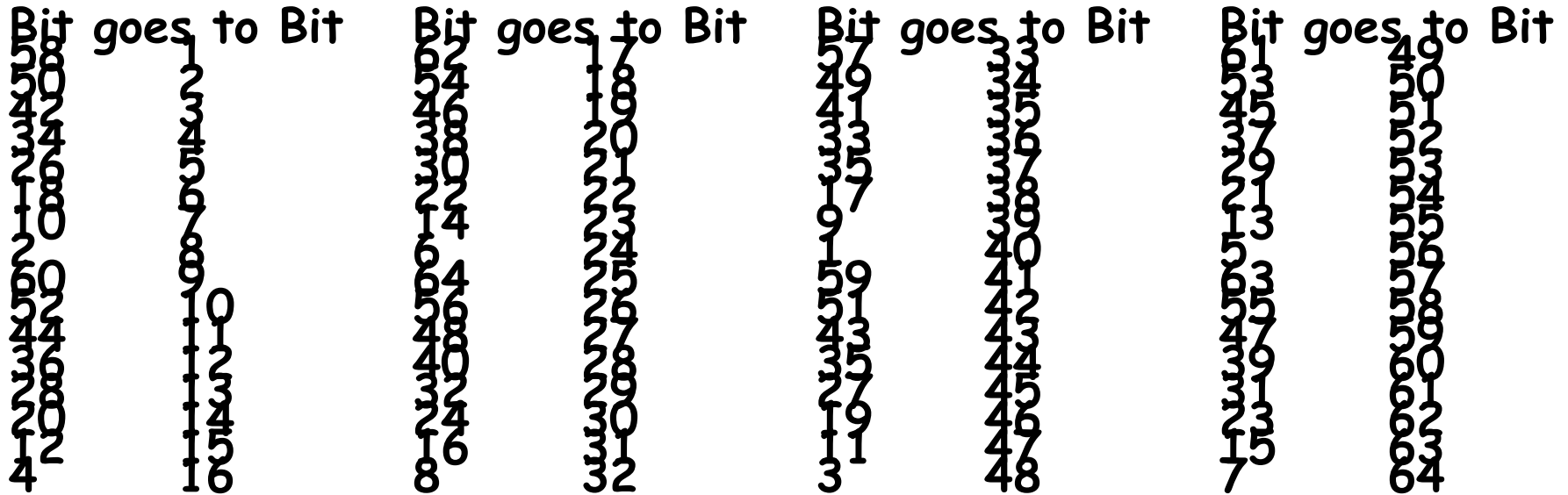
48

49 50 51 52 53 54 55 56 57 58 59 60 61 62 63

64



Initial Permutation



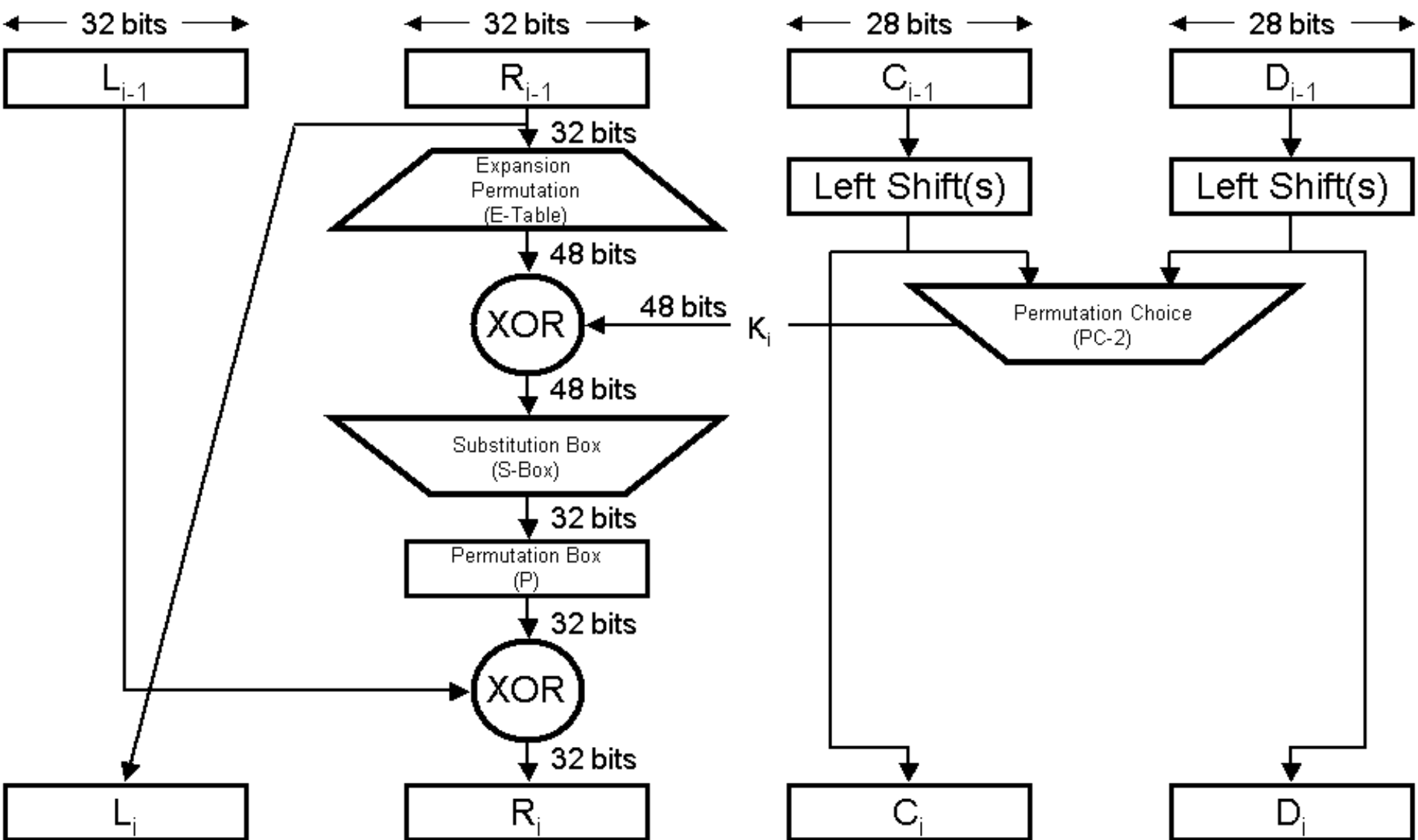
□IP

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

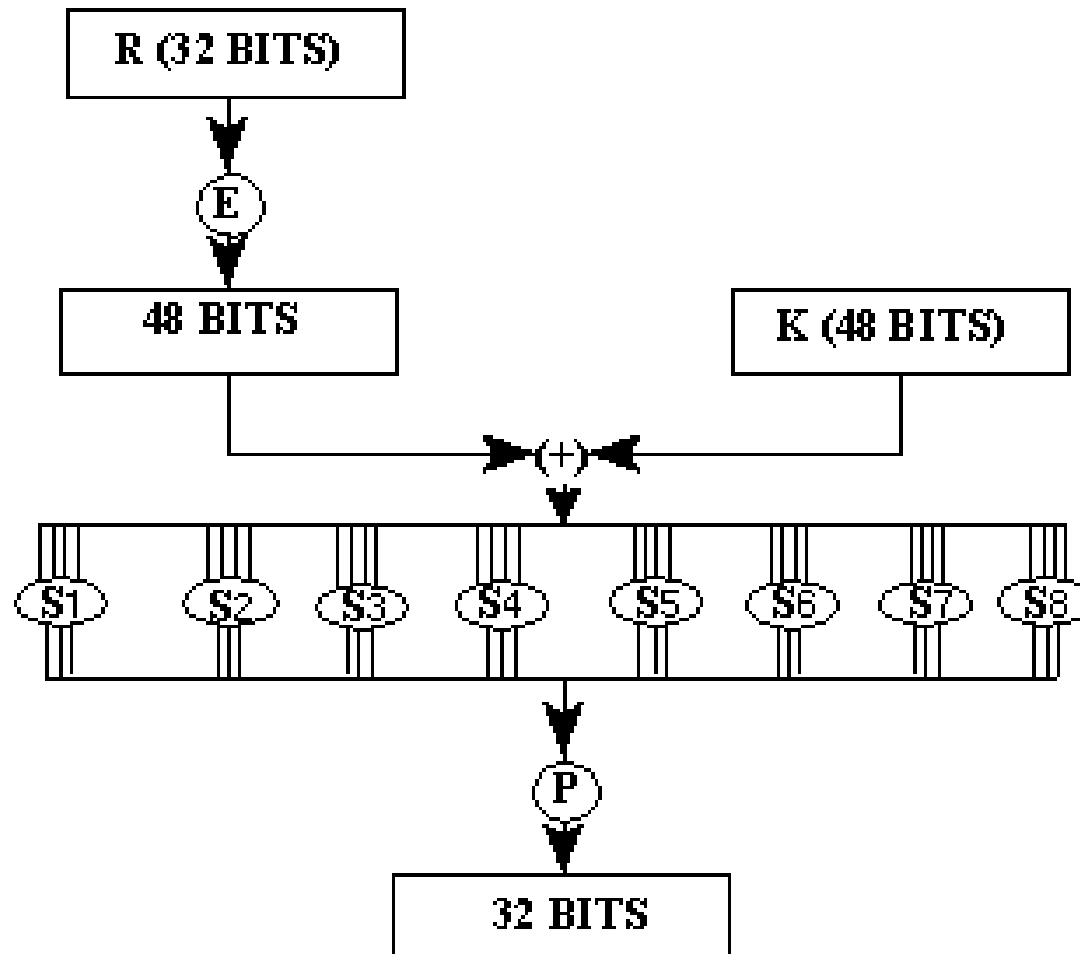
- The permuted input block is then the input to a complex key-dependent computation.
- The output of that computation (*preoutput*) is then subjected to the next permutation which is the inverse of the initial permutation.

DES Numerology

- DES is a Feistel cipher
 - o 64 bit block length
 - o 56 bit key length
 - o 16 rounds
 - o 48 bits of key used in each round (subkey)
- Each round is simple (for a block cipher)
- Security depends primarily on "S-boxes"
 - o Each S-boxes maps 6 bits to 4 bits



Round function for Right Block



DES Expansion Permutation

□ Input 32 bits

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31

□ Output 48 bits

31	0	1	2	3	4	3	4	5	6	7	8
7	8	9	10	11	12	11	12	13	14	15	16
15	16	17	18	19	20	19	20	21	22	23	24
23	24	25	26	27	28	27	28	29	30	31	0

DES S-box

- 8 "substitution boxes" or S-boxes
- Each S-box maps 6 bits to 4 bits
- S-box number 1

input bits (0,5)



input bits (1,2,3,4)

		0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111

00		1110	0100	1101	0001	0010	1111	1011	1000	0011	1010	0110	1100	0101	1001	0000	0111
01		0000	1111	0111	0100	1110	0010	1101	0001	1010	0110	1100	1011	1001	0101	0011	1000
10		0100	0001	1110	1000	1101	0110	0010	1011	1111	1100	1001	0111	0011	1010	0101	0000
11		1111	1100	1000	0010	0100	1001	0001	0111	0101	1011	0011	1110	1010	0000	0110	1101

S-Box with Table entries in decimal

□ What is the output if input is 101000?

S_1															
14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

Output=13 (1101)

$\square S_1$

14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

$\square S_2$

15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9

$\square S_3$

10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12

$\square S_4$

7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14

$\square S_5$

2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3

$\square S_6$

12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13

□S₇

4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12

□S₈

13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

DES P-box

□Input 32 bits

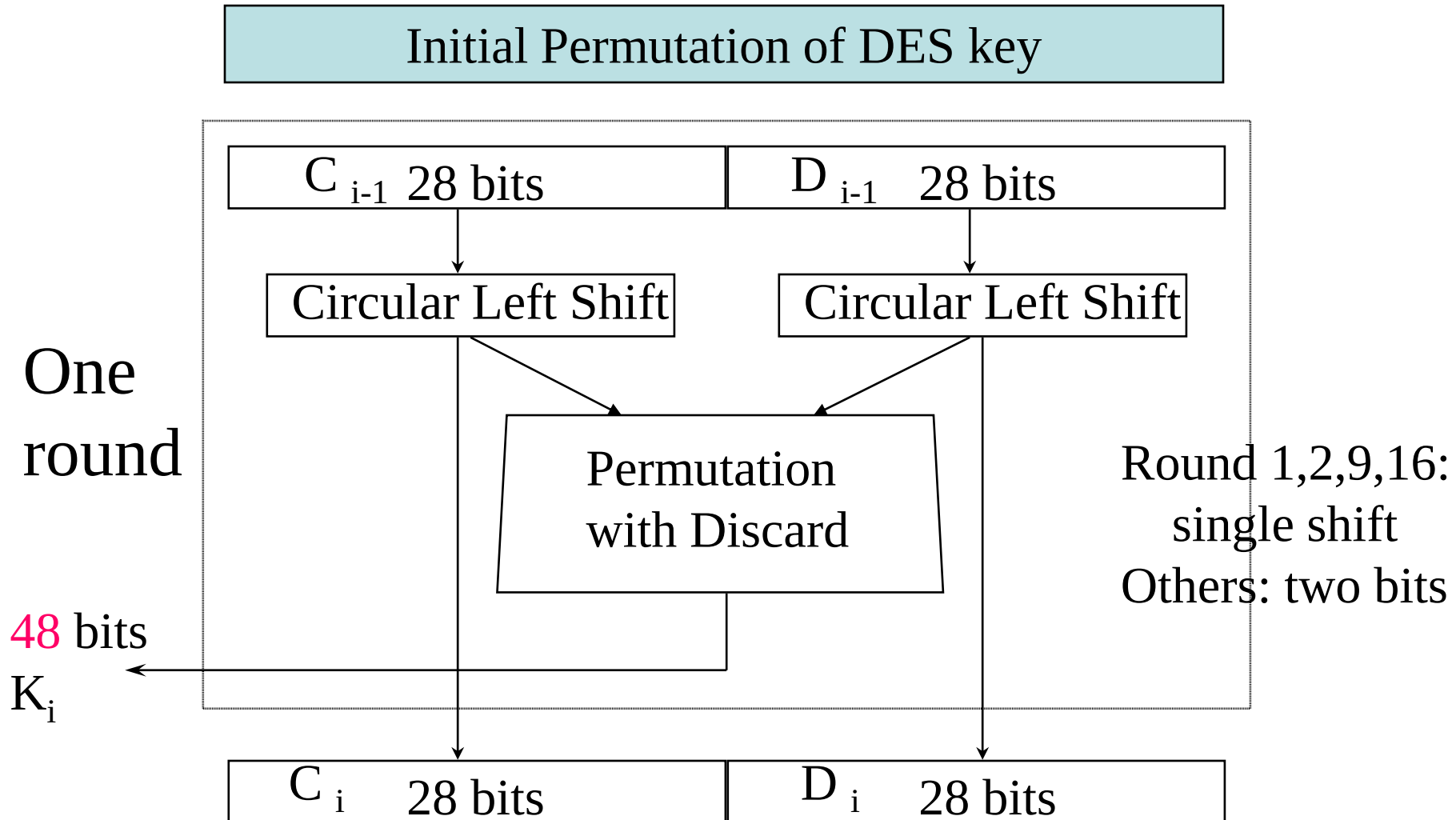
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31

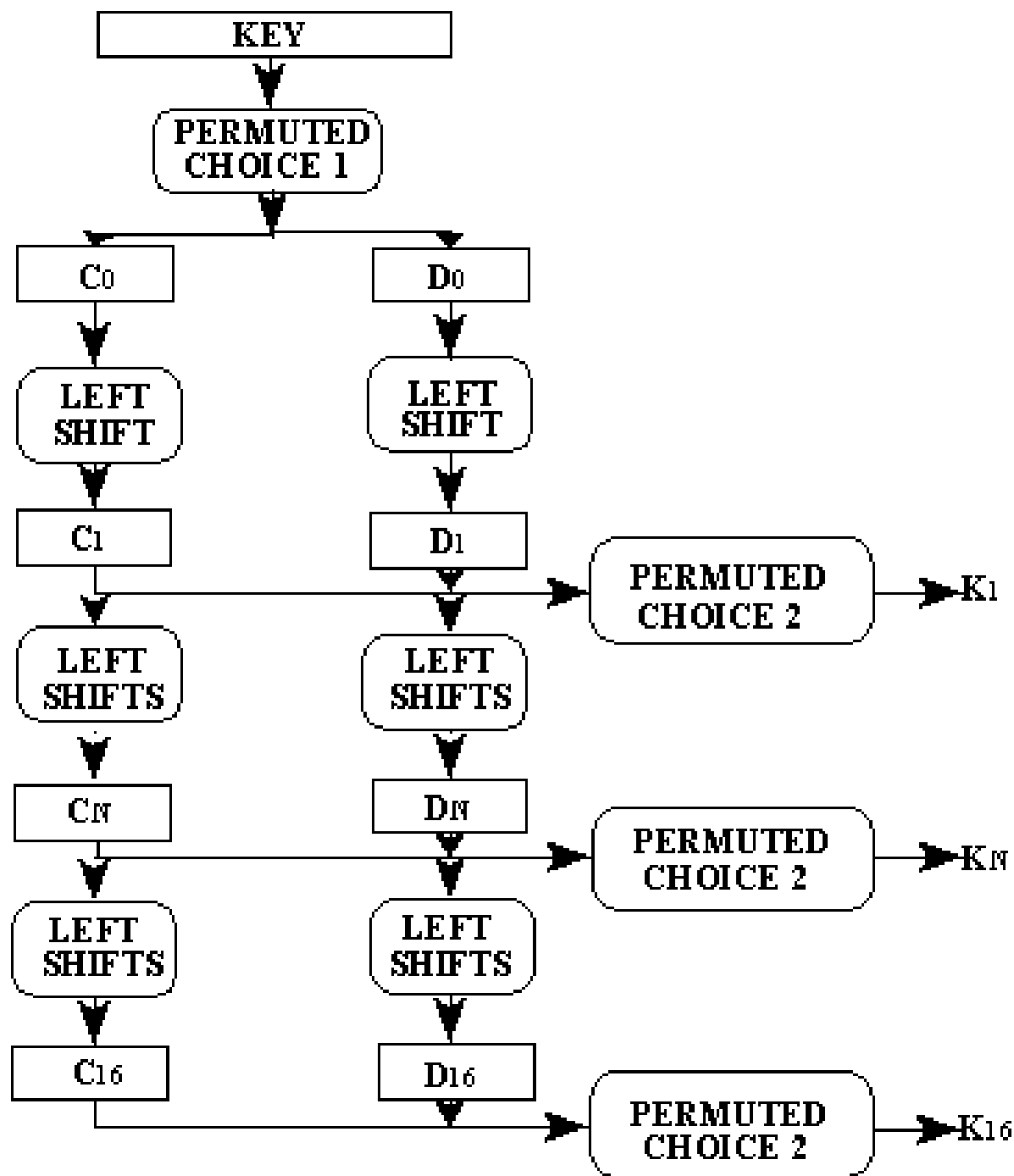
□Output 32 bits

15	6	19	20	28	11	27	16	0	14	22	25	4	17	30	9
1	7	23	13	31	26	2	8	18	12	29	5	21	10	3	24

- The result of p-box is X-OR ed with left block to get next right block.
- Previous right block is next left block
- 16 rounds of key dependent computation is done
- Inverse of initial permutation is performed
- Resulting output is cipher text

Per-Round Key Generation





DES Sub-key Generation

- 56 bit DES key, numbered 0,1,2,...,55 is made to undergo permutation (Permuted Choice 1)
- Permuted key is divided into LK & RK blocks
- Left half key bits, LK (C₀)

49	42	35	28	21	14	7
0	50	43	36	29	22	15
8	1	51	44	37	30	23
16	9	2	52	45	38	31

- Right half key bits, RK (D₀)

55	48	41	34	27	20	13
6	54	47	40	33	26	19
12	5	53	46	39	32	25
18	11	4	24	17	10	3

□ Iteration corresponds to left shifts:

Round	1	2	3	4	5	6	7	8
(r_i)	1	1	2	2	2	2	2	2
	9	10	11	12	13	14	15	16
	1	2	2	2	2	2	2	1

DES Subkey

- For rounds 1, 2, 9 and 16 the shift r_i is 1, and in all other rounds r_i is 2
- Bits 8,17,21,24 of LK omitted after shift by r_i in each round
- Bits 6,9,14,25 of RK omitted after shift by r_i in each round
- **Compression permutation** yields 48 bit sub-key K_i from 56 bits of LK and RK
- **Key schedule** generates sub-key
 - $K(i) = PC2(LS(PC1(K), i))$

DES Subkey

□ For rounds $i=1, 2, \dots, 16$

o Let $LK = (LK \text{ circular shift left by } r_i)$

o Let $RK = (RK \text{ circular shift left by } r_i)$

o Left half of sub-key K_i after Permuted Choice 2

13	16	10	23	0	4	2	27	14	5	20	9
22	18	11	3	25	7	15	6	26	19	12	1

o Right half of sub-key K_i after Permuted Choice 2

12	23	2	8	18	26	1	11	22	16	4	19
15	20	10	27	5	24	17	13	21	7	0	3

□ We sent [DES] S-boxes off to Washington. They came back and were all different

-Alan Konheim

DES Example - Key

K = 581FBC94D3A452EA

X = 3570E2F1BA4682C7

K = (0101 1000 0001 1111 1011 1100 1001 0100
1101 0011 1010 0100 0101 0010 1110 1010)

C0 = (10111100110100 011010010000101)

D0 = (110100100001011 101000001111111)



DES Example - Key

$C1 = (0111\ 1001\ 1010\ 0011\ 0100\ 1000\ 1011)$

$D1 = (1010\ 0100\ 0101\ 1101\ 0000\ 1111\ 1111)$ 

$K1 = (001001\ 111010\ 000101\ 101001$
 $111001\ 011000\ 110111\ 011010)$ 

$C2 = (1111\ 0011\ 0100\ 0110\ 1001\ 0001\ 0110)$
 $D2 = (0100\ 1000\ 1011\ 1010\ 0001\ 1111\ 1111)$ 

$K2 = (110110\ 101001\ 000111\ 011101$
 $110101\ 111011\ 011101\ 001000)$ 

DES Example - Data

K=581FBC94D3A452EA

X=3570E2F1BA4682C7

$X = (x_1, x_2, x_3, \dots, x_{64})$
 $= ($

0011	0101	0111	0000	1110	0010	1111	0001
1011	1010	0100	0110	1000	0010	1100	0111

 $)$

This plaintext X is first subjected to an Initial Permutation - IP which gives

LO	= (	1010	1110	0001	1011	1010	0001	1000	1001)
		^A	^E	¹	^B	^A	¹	⁸	⁹
RO	= (	1101	1100	0001	111	0001	0000	1111	0100)
		^D	^C	¹	^F	¹	⁰	^F	⁴



DES Example - Data

$$E(R0) = \begin{pmatrix} 011011 & 111000 & 000011 & 111110 \\ 100010 & 100001 & 01110 & 101001 \end{pmatrix}$$



$$\begin{aligned} \Gamma 1 &= E(R0) \oplus K1 \\ &= \begin{pmatrix} 010010 & 000010 & 000110 & 010111 \\ 011011 & 111001 & 101001 & 110011 \end{pmatrix} \end{aligned}$$

DES Example - Data

$$S1^{00}(1001) = 1010$$

$$S2^{00}(0001) = 0001$$

$$S3^{00}(0011) = 1110$$

$$S4^{01}(1011) = 1100$$

$$S5^{01}(1101) = S5^1(13) = 9 = 1001$$

$$S6^{11}(1100) = S6^3(12) = 6 = 0110$$

$$S7^{11}(0100) = S7^3(4) = 1 = 0001$$

$$S8^{11}(1001) = S8^3(9) = 12 = 1100$$



DES Example - Data

$B1 = (1010\ 0001\ 1110\ 1100\ 1001\ 0110\ 0001\ 1100)$

$P(B1) = (0010\ 1011\ 1010\ 0001\ 0101\ 0011\ 0110\ 1100)$

$R1 = P(B1) \oplus L0$
 $= (1000\ 0101\ 1011\ 1010\ 1111\ 0010\ 1110\ 0101)$
8 5 B A F 2 E 5



DES Example - Data

$L1 = (1101\ 1100\ 0001\ 1111\ 0001\ 0000\ 1111\ 0100)$
 D C 1 F 1 0 F 4

$E(R1) = (\quad 110000\ 001011\ 110111\ 110101$
 011110 100101 011100 001011)

$\Gamma 2 = E(R1) \oplus K2$
 = (000110 100010 110000 101000
 101011 011110 000001 000011)

DES Example - Data

$S1^{00}(0011)$	=	$S1^1(3)$	= 1	= 0001
$S2^{10}(0001)$	=	$S2^3(1)$	= 14	= 1110
$S3^{10}(1000)$	=	$S3^3(8)$	= 11	= 1011
$S4^{10}(0100)$	=	$S4^3(4)$	= 12	= 1100
$S5^{11}(0101)$	=	$S5^1(5)$	= 14	= 1110
$S6^{00}(1111)$	=	$S6^3(15)$	= 11	= 1011
$S7^{01}(0000)$	=	$S7^3(0)$	= 13	= 1101
$S8^{01}(0001)$	=	$S8^3(1)$	= 15	= 1111



DES Example - Data

$B2 = (0001\ 1110\ 1011\ 1100\ 1110\ 1011\ 1101\ 1111)$

$P(B2) = (0101\ 1111\ 0011\ 1110\ 0011\ 1001\ 1111\ 0111)$

$R2 = P(B2) \oplus L1$
 $= (1000\ 0011\ 0010\ 0001\ 0010\ 1001\ 0000\ 0011)$
 8 3 2 1 2 9 0 3

$L2 = R1 = (1000\ 0101\ 1011\ 1010\ 1111\ 0010\ 1110\ 0101)$
 8 5 B A F 2 E 5

DES Example - Data - Done !

$y = (y_1, y_2, y_3, \dots, y_{64})$
= (1101 0111 0110 1001 1000 0010 0010 0100
0010 1000 0011 1110 0000 1010 1110 1010)
= (D 7 6 9 8 2 2 4 2 8 3 E 0 A E A)

Block Cipher Notation

- P = plaintext block
- C = ciphertext block
- Encrypt P with key K to get ciphertext C
 - $C = E(P, K)$
- Decrypt C with key K to get plaintext P
 - $P = D(C, K)$
- Note that
 - $P = D(E(P, K), K)$ and $C = E(D(C, K), K)$

Triple DES

- Today, 56 bit DES key is too small
- But DES is everywhere: What to do?
- Triple DES or 3DES (112 bit key)
 - $C = E(D(E(P, K_1), K_2), K_1)$
 - $P = D(E(D(C, K_1), K_2), K_1)$
- Why use Encrypt-Decrypt-Encrypt (EDE) with 2 keys?
 - Backward compatible: $E(D(E(P, K), K), K) = E(P, K)$
 - And 112 bits is enough

3DES

- Why not $C = E(E(P, K), K)$?
 - Still just 56 bit key
- Why not $C = E(E(P, K_1), K_2)$?
- A (semi-practical) known plaintext attack
 - Precompute table of $E(P, K_1)$ for every possible key K_1 (resulting table has 2^{56} entries)
 - Then for each possible K_2 compute $D(C, K_2)$ until a match in table is found
 - When match is found, have $E(P, K_1) = D(C, K_2)$
 - Result is keys: $C = E(E(P, K_1), K_2)$

Advanced Encryption Standard

- ❑ Replacement for DES
- ❑ AES competition (late 90's)
 - o NSA openly involved
 - o Transparent process
 - o Many strong algorithms proposed
 - o Rijndael Algorithm ultimately selected
 - Pronounced like "Rain Doll" or "Rhine Doll"
 - o Developed by Joan Daemen & Vincent Rijmen
- ❑ Iterated block cipher (like DES)
- ❑ Not a Feistel cipher (unlike DES)

AES Overview

- ❑ **Block size:** 128, 192 or 256 bits
- ❑ **Key length:** 128, 192 or 256 bits
(independent of block size)
- ❑ 10 to 14 rounds (depends on key length)
- ❑ Each round uses 4 functions (in 3 "layers")
 - oByteSub (nonlinear layer)
 - oShiftRow (linear mixing layer)
 - oMixColumn (nonlinear layer)
 - oAddRoundKey (key addition layer)

□ Data encrypted with 256 bit AES key will be 'secure for ever' regardless of any conceivable advances in computing technology

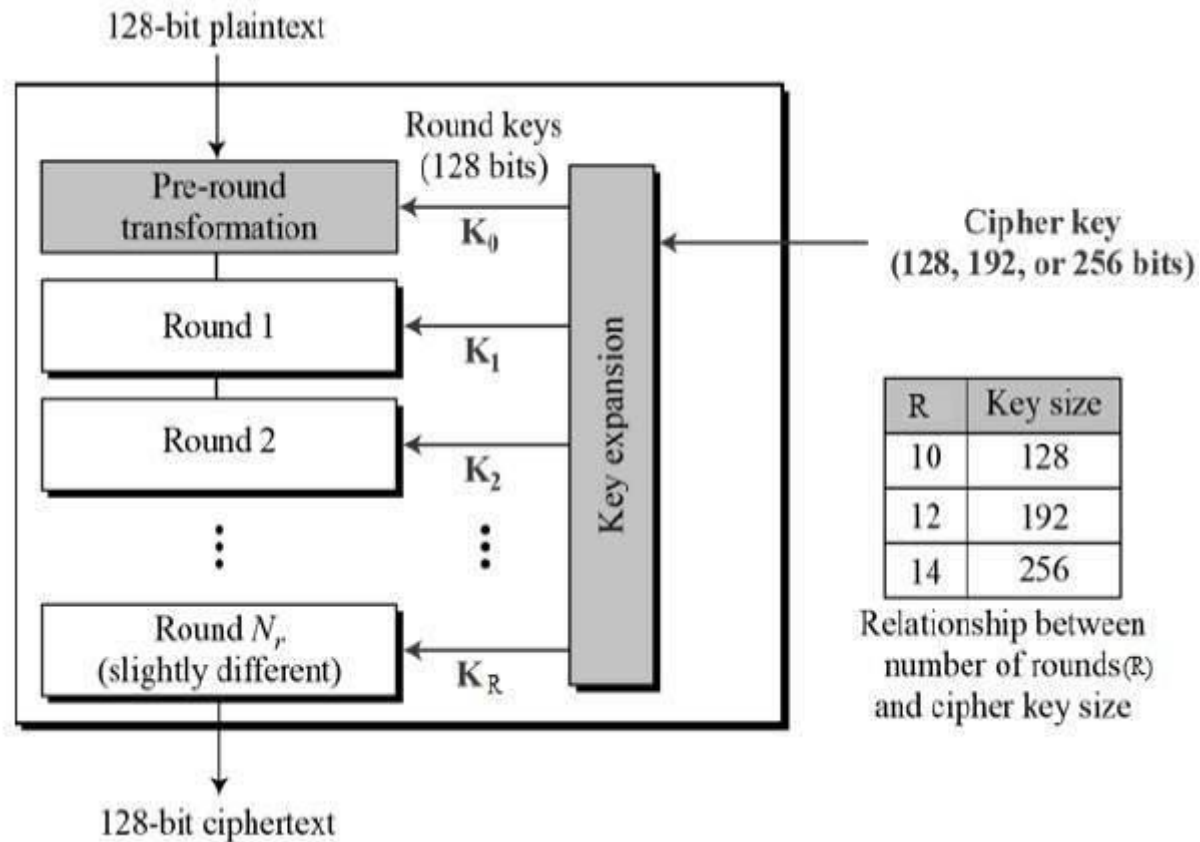
--- Shamir

□ Even if NSA manages to build a machine with 1 billion parallel processors, each being able to evaluate one key per picosecond, it would take such a machine about 10^{10} years to search the key space. By then the sun would have burned out, so the folks then present will have to read the results with candle light

---Andrew

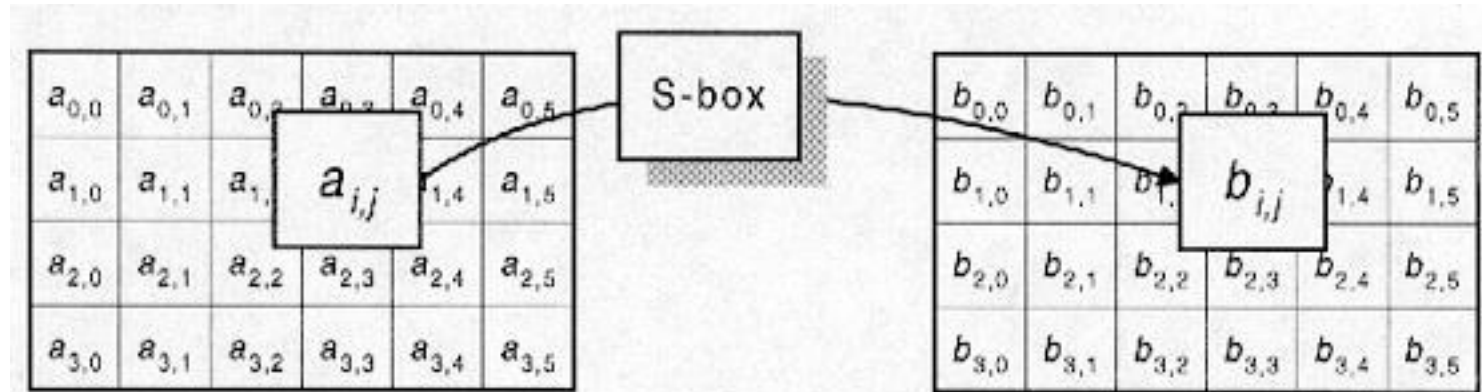
Tanenbaum

AES



AES ByteSub

- Assume 192 bit block, 4x6 bytes



- ByteSub is AES's "S-box"
- Can be viewed as nonlinear (but invertible) composition of two math operations

AES "S-box"

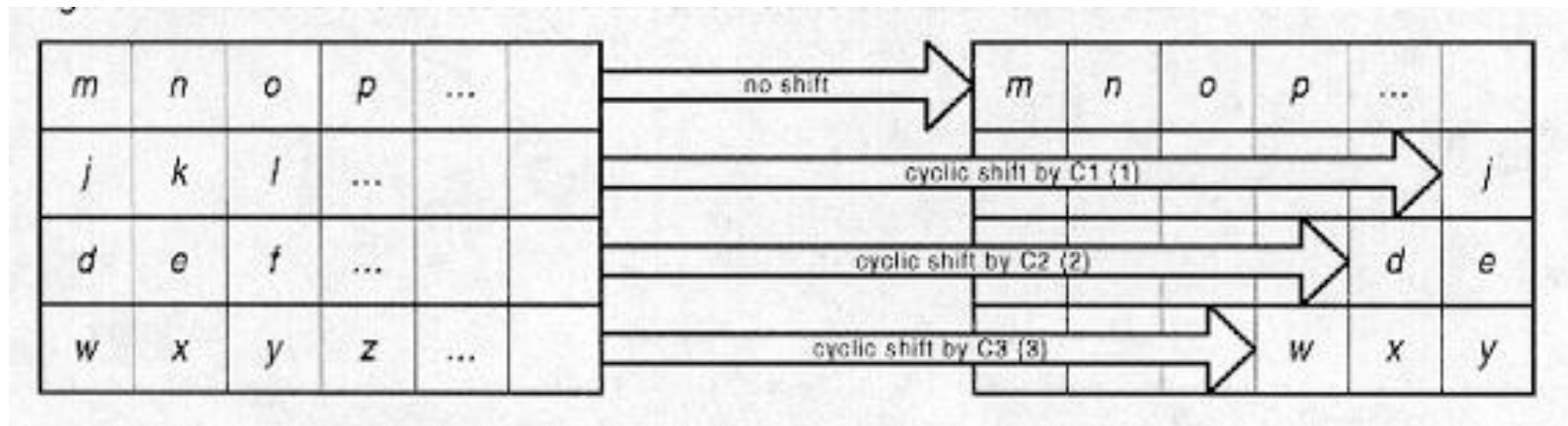
Last 4 bits of input

First 4
bits of
input

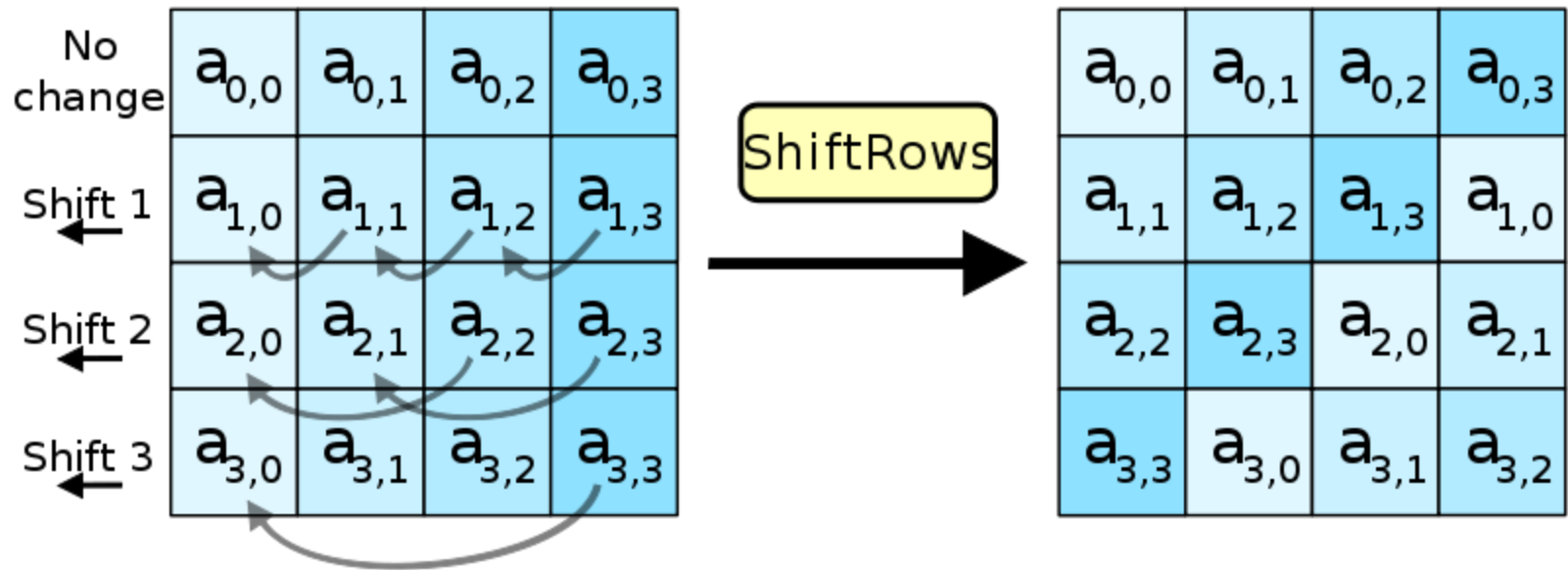
	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
0	63	7c	77	7b	f2	6b	6f	c5	30	01	67	2b	fe	d7	ab	76
1	ca	82	c9	7d	fa	59	47	f0	ad	d4	a2	af	9c	a4	72	c0
2	b7	fd	93	26	36	3f	f7	cc	34	a5	e5	f1	71	d8	31	15
3	04	c7	23	c3	18	96	05	9a	07	12	80	e2	eb	27	b2	75
4	09	83	2c	1a	1b	6e	5a	a0	52	3b	d6	b3	29	e3	2f	84
5	53	d1	00	ed	20	fc	b1	5b	6a	cb	be	39	4a	4c	58	cf
6	d0	ef	aa	fb	43	4d	33	85	45	f9	02	7f	50	3c	9f	a8
7	51	a3	40	8f	92	9d	38	f5	bc	b6	da	21	10	ff	f3	d2
8	cd	0c	13	ec	5f	97	44	17	c4	a7	7e	3d	64	5d	19	73
9	60	81	4f	dc	22	2a	90	88	46	ee	b8	14	de	5e	0b	db
a	e0	32	3a	0a	49	06	24	5c	c2	d3	ac	62	91	95	e4	79
b	e7	c8	37	6d	8d	d5	4e	a9	6c	56	f4	ea	65	7a	ae	08
c	ba	78	25	2e	1c	a6	b4	c6	e8	dd	74	1f	4b	bd	8b	8a
d	70	3e	b5	66	48	03	f6	0e	61	35	57	b9	86	c1	1d	9e
e	e1	f8	98	11	69	d9	8e	94	9b	1e	87	e9	ce	55	28	df
f	8c	a1	89	0d	bf	e6	42	68	41	99	2d	0f	b0	54	bb	16

AES ShiftRow

□Cyclic shift rows

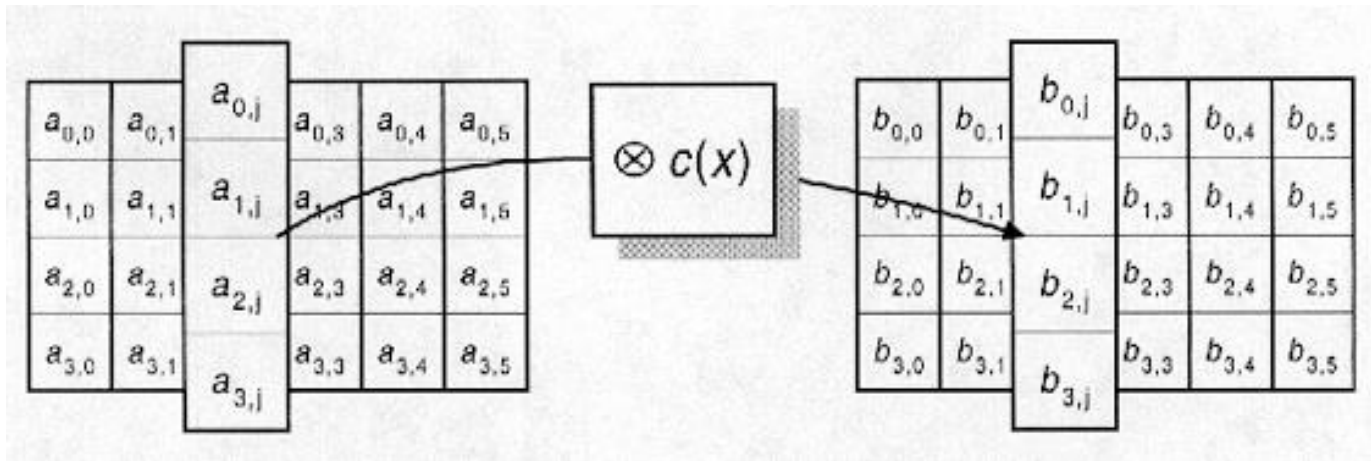


AES ShiftRow



AES MixColumn

- Nonlinear, invertible operation applied to each column



- Implemented as a (big) lookup table

□ During this operation, each column is multiplied by the known constant matrix. For the 128 bit key the constant matrix is

$$\begin{bmatrix} 2 & 3 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 3 \\ 3 & 1 & 1 & 2 \end{bmatrix}.$$

AES AddRoundKey

□ XOR subkey with block

$$\begin{bmatrix} a_{00} & a_{01} & a_{02} & a_{03} & a_{04} & a_{05} \\ a_{10} & a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{20} & a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{30} & a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \end{bmatrix} \oplus \begin{bmatrix} k_{00} & k_{01} & k_{02} & k_{03} & k_{04} & k_{05} \\ k_{10} & k_{11} & k_{12} & k_{13} & k_{14} & k_{15} \\ k_{20} & k_{21} & k_{22} & k_{23} & k_{24} & k_{25} \\ k_{30} & k_{31} & k_{32} & k_{33} & k_{34} & k_{35} \end{bmatrix} = \begin{bmatrix} b_{00} & b_{01} & b_{02} & b_{03} & b_{04} & b_{05} \\ b_{10} & b_{11} & b_{12} & b_{13} & b_{14} & b_{15} \\ b_{20} & b_{21} & b_{22} & b_{23} & b_{24} & b_{25} \\ b_{30} & b_{31} & b_{32} & b_{33} & b_{34} & b_{35} \end{bmatrix}$$

Block Subkey

□ RoundKey (subkey) determined by **key schedule algorithm**

AES Decryption

- ❑ To decrypt, process must be invertible
- ❑ Inverse of MixAddRoundKey is easy, since $\oplus$ is its own inverse
- ❑ MixColumn is invertible (inverse is also implemented as a lookup table)
- ❑ Inverse of ShiftRow is easy (cyclic shift the other direction)
- ❑ ByteSub is invertible (inverse is also implemented as a lookup table)

AES Example

Plaintext – Two One Nine Two

T	w	o		O	n	e		N	i	n	e		T	w	o
54	77	6F	20	4F	6E	65	20	43	69	6E	25	20	54	77	6F

Plaintext in Hex Format

54 77 6F 20 4F 6E 65 20 43 69 6E 25 20 54 77 6F

Encryption Key – Thats my Kung Fu

T	h	a	t	s		m	y		K	u	n	g		F	u
54	68	61	74	73	20	6D	79	20	4B	75	6E	67	20	46	75

Encryption Key in Hex Format

54 68 61 74 73 20 6D 79 20 4B 75 6E 67 20 46 75

Keys

Keys generated for every round

- Round 0: 54 68 61 74 73 20 6D 79 20 4B 75 6E 67 20 46 75
- Round 1: E2 32 FC F1 91 12 91 88 B1 59 E4 E6 D6 79 A2 93
- Round 2: 56 08 20 07 C7 1A B1 8F 76 43 55 69 A0 3A F7 FA
- Round 3: D2 60 0D E7 15 7A BC 68 63 39 E9 01 C3 03 1E FB
- Round 4: A1 12 02 C9 B4 68 BE A1 D7 51 57 A0 14 52 49 5B
- Round 5: B1 29 3B 33 05 41 85 92 D2 10 D2 32 C6 42 9B 69
- Round 6: BD 3D C2 B7 B8 7C 47 15 6A 6C 95 27 AC 2E 0E 4E
- Round 7: CC 96 ED 16 74 EA AA 03 1E 86 3F 24 B2 A8 31 6A
- Round 8: 8E 51 EF 21 FA BB 45 22 E4 3D 7A 06 56 95 4B 6C
- Round 9: BF E2 BF 90 45 59 FA B2 A1 64 80 B4 F7 F1 CB D8
- Round 10: 28 FD DE F8 6D A4 24 4A CC C0 A4 FE 3B 31 6F 26

Add Round Key:

54	4F	4E	20
77	6E	69	54
6F	65	6E	77
20	20	65	6F

Plaintext



XOR

54	73	20	67
68	20	4B	20
61	6D	75	46
74	79	6E	75

Round 0 Key

00	3C	63	47
1F	4E	22	74
0E	08	1B	31
54	59	0B	1A

New State Array

Sub-Bytes

63	EB	9F	A0
C0	2F	93	92
AB	30	AF	C7
20	CB	2B	A2

New State Array

Shift Rows

63	EB	9F	A0
C0	2F	93	92
AB	30	AF	C7
20	CB	2B	A2

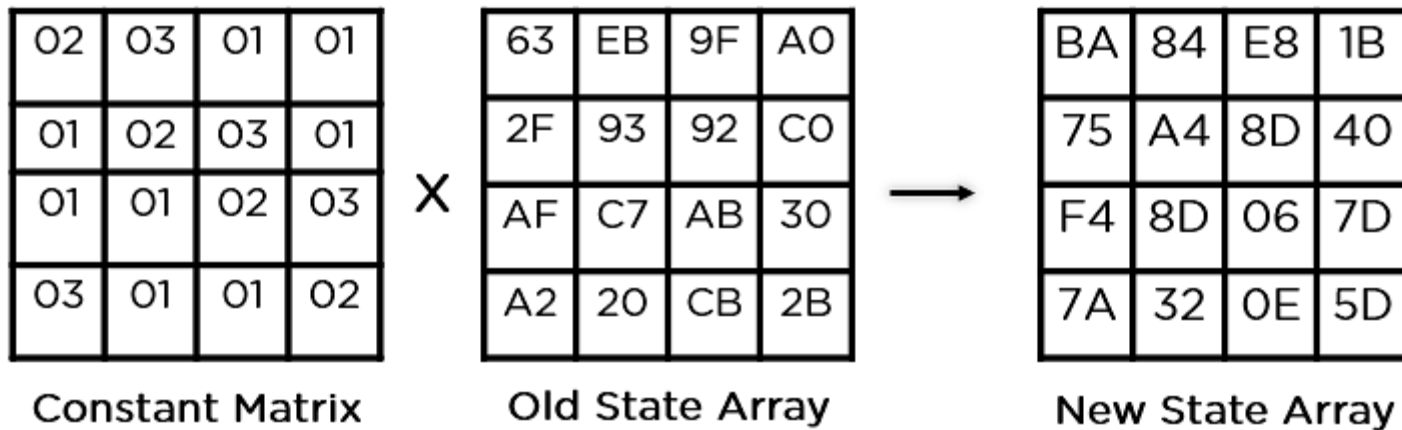
Old State Array



63	EB	9F	A0
2F	93	92	C0
AF	C7	AB	30
A2	20	CB	2B

New State Array

Mix Columns:



Add Round Key:

BA	84	E8	1B
75	A4	8D	40
F4	8D	06	7D
7A	32	0E	5D

Old State Array



XOR

E2	91	B1	D6
32	12	59	79
FC	91	E4	A2
F1	88	E6	93

Round 1 Key

58	15	59	CD
47	B6	D4	39
08	1C	E2	DF
8B	BA	E8	CE

New State Array

Ciphertext for this particular round. This becomes the input for the next round

Final State Array after Round 10

29	57	40	1A
C3	14	22	02
50	20	99	D7
5F	F6	B3	3A

AES Final Output

29 C3 50 5F 57 14 20 F6 40 22 99 B3 1A 02 D7 3A



Ciphertext